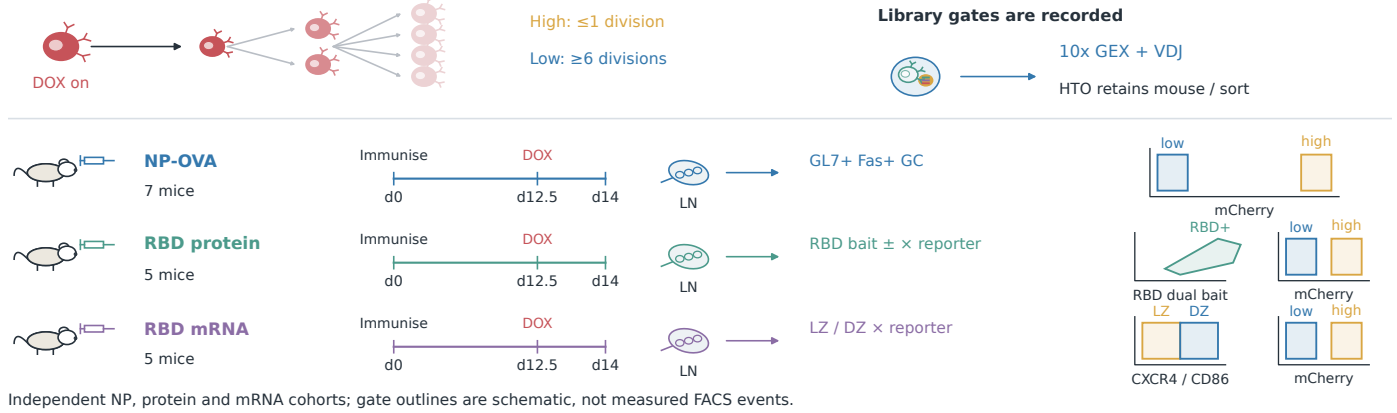


Figure 2 | Model-antigen GC state nodes and independent reporter measurements

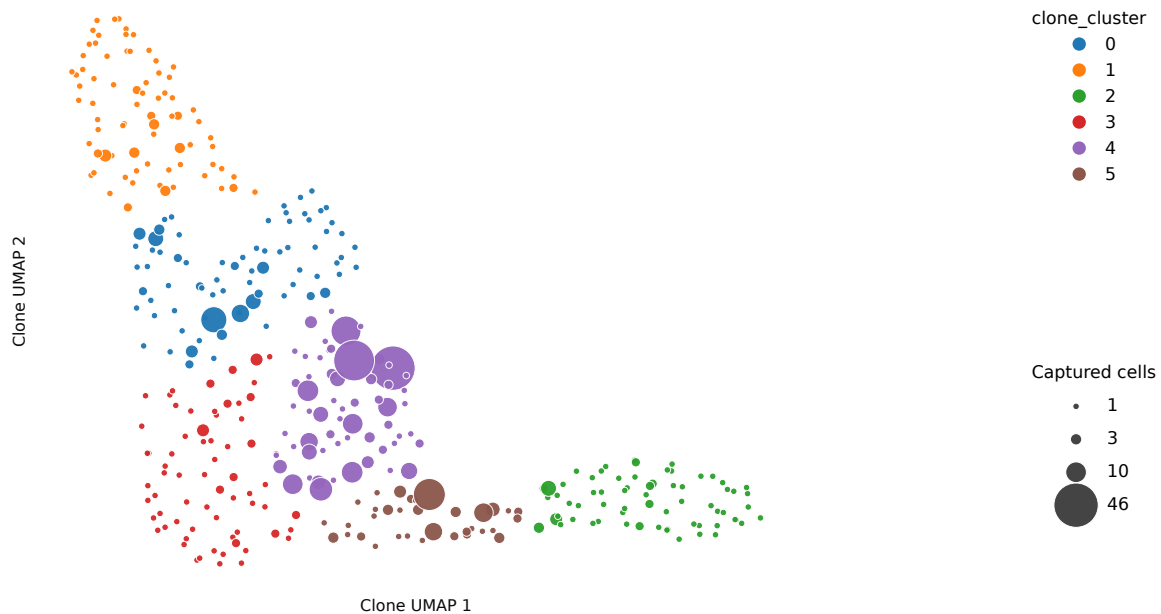
Reconstructed clone embedding; Leiden colours, captured-cell sizes and measured GC-state context.

A Independent NP-OVA and RBD reporter designs

H2B-mCherry reporter: stop synthesis, read 36 h of division history

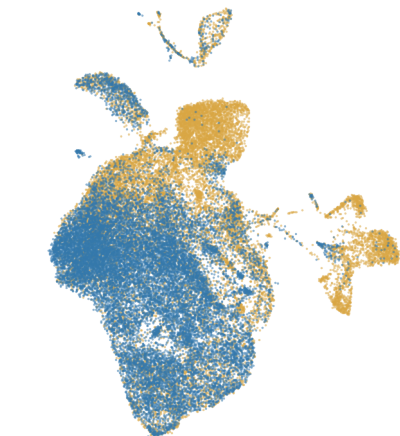


B NP-OVA day 14: clone embedding and Leiden reclustering

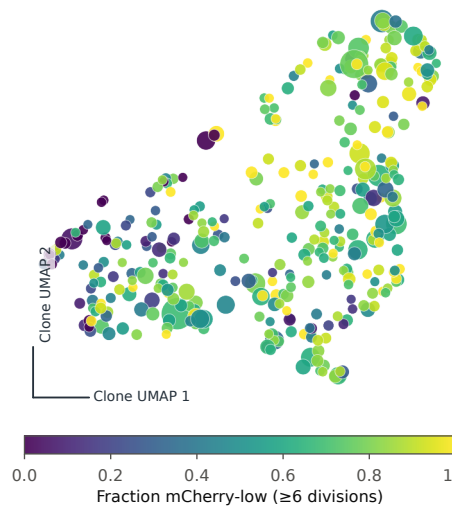


GSE246382: 377 same-mouse V-D-J receptor groups / 762 cells; ≥ 1 cell/group. Measured context: S8B-E.

C RBD: measured division history per cell (mCherry-low = ≥ 6 divisions / 36 h)



D RBD: division history per clone family (fraction of family mCherry-low)



E RBD: V-region mutation history on the same clone map

